Redis Configuration Guide for GEMA Backend

Critical: BullMQ Eviction Policy Requirement

BullMQ (background job queue system) requires Redis to use the `noeviction` policy. Using other policies like `volatile-lru` or `allkeys-lru` can cause job data corruption and worker failures.

Current Issue

If you're seeing these errors in production:

```
IMPORTANT! Eviction policy is volatile-lru. It should be "noeviction"
Stream isn't writeable and enableOfflineQueue options is false
```

Your Redis server is configured with the wrong eviction policy.

Quick Fix (Immediate - No Restart Required)

Connect to your Redis server and run:

```
# Connect to Redis
redis-cli -h <REDIS_HOST> -p <REDIS_PORT> -a <REDIS_PASSWORD>

# Check current policy
CONFIG GET maxmemory-policy
# Output: 1) "maxmemory-policy"  2) "volatile-lru"  <-- Wrong!

# Fix: Set to noeviction
CONFIG SET maxmemory-policy noeviction

# Verify the change
CONFIG GET maxmemory-policy
# Output: 1) "maxmemory-policy"  2) "noeviction"  <-- Correct!
```

This change takes effect immediately without restarting Redis.

Make Fix Permanent

To ensure the fix persists after Redis restarts:

Option 1: Edit redis.conf directly

1. Find your Redis configuration file:

   ```
   # Common locations:
   # /etc/redis/redis.conf
   # /etc/redis.conf
   # /usr/local/etc/redis.conf

   # Or find it:
   redis-cli CONFIG GET dir
   redis-cli CONFIG GET config_file
   ```

2. Edit the file:

   ```
   sudo nano /etc/redis/redis.conf
   ```

3. Find and change the line:

   ```
   # FROM:
   maxmemory-policy volatile-lru

   # TO:
   maxmemory-policy noeviction
   ```

4. Save and restart Redis:

```
sudo systemctl restart redis
# OR
sudo service redis restart
```

Option 2: Use Redis CONFIG REWRITE

```
redis-cli -h <REDIS_HOST> -p <REDIS_PORT> -a <REDIS_PASSWORD>
CONFIG SET maxmemory-policy noeviction
CONFIG REWRITE
```

This automatically updates the redis.conf file (if Redis has write permissions).

Using Managed Redis Services

Redis Cloud (RedisLabs)

1. Log into Redis Cloud console
2. Select your database
3. Go to Configuration → Advanced Options
4. Set "Eviction Policy" to "noeviction"
5. Save changes

AWS ElastiCache

1. Open ElastiCache console
2. Select your Redis cluster
3. Click "Modify"
4. Find "Parameter Group"
5. Create/modify parameter group with `maxmemory-policy = noeviction`
6. Apply changes

DigitalOcean Managed Redis

1. Open DigitalOcean control panel
2. Select your Redis cluster
3. Go to Settings → Advanced
4. Set "Maxmemory Policy" to "noeviction"
5. Save

Upstash

1. Log into Upstash console
2. Select your database
3. Go to Settings
4. Set "Eviction Policy" to "noeviction"

Why noeviction?

BullMQ stores job data, state, and metadata in Redis. With eviction policies like `volatile-lru` or `allkeys-lru`:

- Redis may evict job data when memory is low
- Evicted jobs are permanently lost
- Queue state becomes corrupted
- Workers fail with "Stream isn't writeable" errors

With `noeviction`:

- Redis returns errors when memory is full instead of evicting data
- Job data integrity is guaranteed
- Application can handle memory errors gracefully

Memory Management with noeviction

If you're concerned about memory usage:

1. **Set maxmemory limit** (recommended):

   ```
   CONFIG SET maxmemory 256mb  # Adjust based on your needs
   ```

2. **Configure job retention** (already optimized in code):

- Completed jobs: 1 hour retention
- Failed jobs: 24 hour retention
- Max 100 recent completed jobs kept

3. **Monitor memory usage**:

```
redis-cli INFO memory
```

4. **Manual cleanup if needed**:

```
# Clean old completed jobs
redis-cli --scan --pattern "bull:*:completed" | xargs redis-cli DEL
```

Verification

After applying the fix, verify in your application logs:

```
# You should see:
BullMQ Redis: Connected and ready

# You should NOT see:
IMPORTANT! Eviction policy is volatile-lru. It should be "noeviction"
```

Environment Variables

Ensure these are set in your `.env` file:

```
REDIS_HOST=localhost            # Your Redis host
REDIS_PORT=6379                 # Your Redis port
REDIS_PASSWORD=                 # Your Redis password (if any)
REDIS_DB=0                      # Database number
REDIS_TLS=false                 # Set to true for TLS connections
DISABLE_REDIS=false             # Keep false for production
```

Troubleshooting

Still seeing errors after fix?

1. **Verify config applied**:

```
redis-cli CONFIG GET maxmemory-policy
```

2. **Restart workers**:

```
pm2 restart gema-backend
pm2 restart gema-worker
```

3. **Check Redis connectivity**:

```
redis-cli -h <HOST> -p <PORT> PING
# Should return: PONG
```

4. **Check Redis logs**:

```
# Ubuntu/Debian
sudo tail -f /var/log/redis/redis-server.log

# CentOS/RHEL
sudo tail -f /var/log/redis/redis.log
```

Connection issues?

Check backend logs for:

```
BullMQ Redis: Reconnecting... Attempt X
```

If reconnection attempts exceed 50, there's a persistent connection issue. Check:

- Redis server is running
- Firewall allows connections
- Credentials are correct

- Network connectivity

Additional Resources

- [BullMQ Documentation (https://docs.bullmq.io/)](https://docs.bullmq.io/)
- [Redis Eviction Policies (https://redis.io/docs/manual/eviction/)](https://redis.io/docs/manual/eviction/)
- [Redis Configuration (https://redis.io/docs/management/config/)](https://redis.io/docs/management/config/)

Need Help?

Contact your system administrator or check:

- Backend logs: `pm2 logs gema-backend`
- Worker logs: `pm2 logs gema-worker`
- Redis logs: `/var/log/redis/`